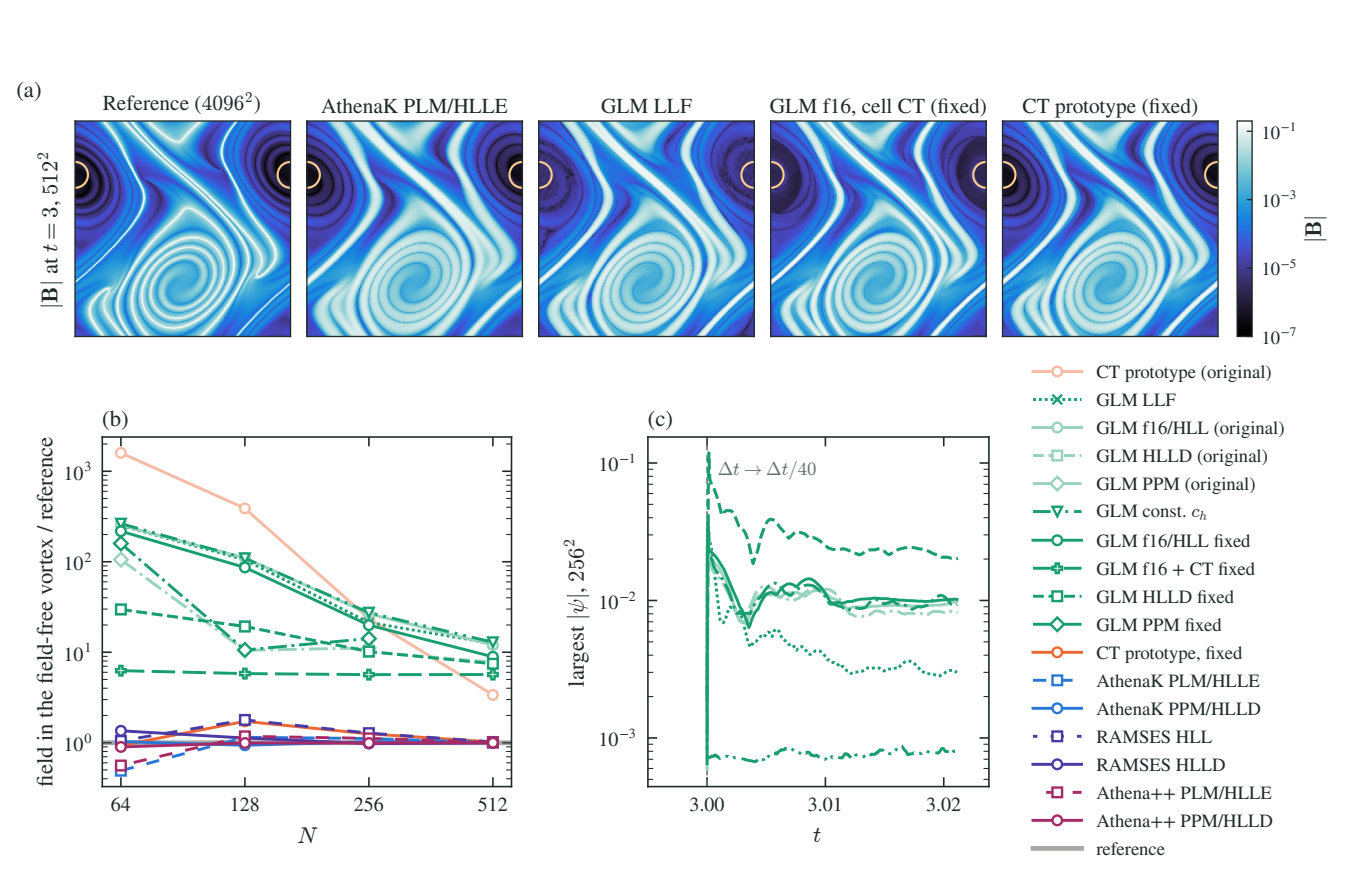

(a)
Reference (4096^2)
AthenaK PLM/HLLE
GLM LLF
GLM f16, cell CT (fixed)
CT prototype (fixed)
$|\mathbf{B}|$ at $t = 3,512^2$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
$|\mathbf{B}|$
(b)
field in the field-free vortex / reference
10^3
10^2
10^1
10^0
64
128
256
512
N
(c)
largest $|\psi|$, 256^2
$\Delta t \to \Delta t/40$
10^{-1}
10^{-2}
10^{-3}
3.00
3.01
3.02
t
CT prototype (original)
GLM LLF
GLM f16/HLL (original)
GLM HLLD (original)
GLM PPM (original)
GLM const. c_h
GLM f16/HLL fixed
GLM f16 + CT fixed
GLM HLLD fixed
GLM PPM fixed
CT prototype, fixed
AthenaK PLM/HLLE
AthenaK PPM/HLLD
RAMSES HLL
RAMSES HLLD
Athena++ PLM/HLLE
Athena++ PPM/HLLD
reference